The Clebsch hexagon code: rigidity of a conic syndrome locus

An MDS code over $\mathbb{F}_{11}$ whose projective deep-hole directions form a conic

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Working draft — July 2026

Abstract

Among all six-arcs of the projective plane $\mathrm{PG}(2,11)$, exactly one — up to the action of $\mathrm{PGL}(3,11)$ — has the property that the projective maximum-distance syndrome directions of its associated MDS code lie on a conic. That arc is the *Clebsch hexagon*, an icosahedral six-arc with full projective stabilizer A_5, and the conic condition alone recovers that stabilizer. The arc lies on no conic, so its $[6,3,4]_{11}$ code is not monomially equivalent to a GRS code and has covering radius three. Its twelve projective deep-hole syndrome directions are exactly the $\mathbb{F}_{11}$-rational points of that conic; they lift to 120 deep-hole cosets, 2400 minimum-weight leaders, and 159720 received-word deep holes. The 120 cosets form one monomial orbit, and the 159720 received words form one orbit after adjoining translations by codewords. This identification is recorded with its certificate in a companion paper and taken here as the starting point. We prove this rigidity as a five-way equivalence, prove a sharp global nearest-conic gap of twelve, decompose the 252 one-point perturbations into eight A_5-orbits, exhibit a canonical automorphism-invariant unordered bipartition of the twenty weight-three support types (and hence a $10+10$ split in every deep-hole coset), and show that $q=11$ is the only prime power at which the conic-filling syndrome locus can occur at all: a chord-multiplicity identity reduces every characteristic to $q \in \{4,5,9,11\}$, and the $q=9$ residue is excluded by the exact-distance-two clique number of the Sylvester graph. The underlying six-arc census is classical; we recompute its extension-count spectrum and supply the coding-theoretic reading, but claim no priority on the raw numbers themselves.

1 Introduction

Let C be a linear $[n,k,d]_q$ code with covering radius ρ and full-rank parity-check matrix H. For a received word v, write $\sigma(v) = Hv^{\mathsf{T}}$ and

$$w(s) = \min\{\mathrm{wt}(e) : He^{\mathsf{T}} = s\}.$$

Then $d(v,C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ. The syndrome s specifies one coset of C, whose coset leaders are its weight-$w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\mathrm{proj}}(C) = \{[s] \in \mathrm{PG}(n-k-1,q) : w(s) = \rho\}.$$

Thus one projective direction represents $q-1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n-k$, the parity-check columns form a projective arc in $\mathrm{PG}(r-1,q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r-1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. This arc–coset dictionary, in various forms, underlies the classical theory of complete arcs and saturating sets and has been made precise and general for MDS codes by Davydov, Marcugini, and Pambianco [7], whose Theorem 6.3 gives the exact leader-count formula we use in Section 3. For plane parity-check systems — equivalently redundancy-three $[n, n-3, 4]_q$ MDS codes — the same dictionary is implicit already in the classical arc/covering-radius literature; see Hirschfeld [17]. We write $\mathcal{U}(A)$ for this projective extension locus, the directions off A and off all its secants. It is the weight-three syndrome-direction locus. When $\mathcal{U}(A) \neq \emptyset$ the associated redundancy-three code has $\rho = 3$ and $\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C)$; when $\mathcal{U}(A) = \emptyset$ the covering radius is at most two, and its deep-hole directions are therefore not described by $\mathcal{U}(A)$. For the standard (affine) Reed–Solomon setting, determining deep holes is a long-studied algorithmic and structural question; Guruswami and Vardy [16] initiated its decision-complexity study, showing maximum-likelihood decoding of Reed–Solomon codes to be NP-hard by essentially proving that deciding whether a received word is a deep hole is co-NP-complete, and Cheng and Murray [6] gave a substantially simpler proof of that same co-NP-completeness result and extended the study to standard (full-evaluation-set) Reed–Solomon codes. Zhang, Wan, and Kaipa [28] determine the deep holes of *projective* Reed–Solomon codes completely, identifying them with tangent lines to, and a quadratic-extension family over, the underlying normal rational curve (their Theorems I.4–I.7). Both of these are for the generalized Reed–Solomon (GRS) case, where the arc lies on a normal rational curve; the present code does not.

The object of this paper is a single, small, and highly symmetric exception to that GRS world: a $[6, 3, 4]_{11}$ MDS code not monomially equivalent to a GRS code, whose covering radius is three and whose projective deep-hole syndrome locus — the directions missed by the arc's fifteen secants — coincides exactly with the $\mathbb{F}_{11}$-points of a single nonsingular conic. The arc realizing this code is the *Clebsch hexagon*: six points of $\mathrm{PG}(2, 11)$. Its real-projective antecedent goes back to Clebsch [3, p. 336]; Edge constructed the finite-plane configuration, named it a Clebsch hexagon, and determined its order-60 stabilizer [4, §§29–32]. Dye developed its synthetic geometry over a general field [14], and Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [25].

That such an arc exists, and that its projective deep-hole syndrome locus is a conic, is by itself a finite check — an instance of the DMP dictionary applied to one known arc. What we add is that this instance is not an accident of the arc chosen: among *all* six-arcs of $\mathrm{PG}(2, 11)$ — fifteen classes under $\mathrm{PGL}(3, 11)$, which our search meets as 1548 frame-normalized representatives — the Clebsch hexagon is the unique class whose projective deep-hole syndrome locus lies on any conic at all, degenerate or not, and this single containment condition already forces the stabilizer to contain A_5. We state this as a rigidity theorem (Section 4), quantify it by a global nearest-conic gap and an orbit-resolved local perturbation theorem, exhibit an invariant chirality bipartition of the weight-three support types and its induced per-coset leader split (Section 5), and prove that $q = 11$ is the only prime power at which the conic-filling extension locus can occur (Section 6). The proof combines a universal chord-defect identity with the Sylvester graph on the internal points of a conic in $\mathrm{PG}(2, 9)$; it assumes no symmetry of the arc. The previous nine-field census is retained as independent exact verification.

What is new and what is not. The six-arc itself, its real-projective ancestry, and its finite conic geometry are classical: Clebsch [3], Edge [4], and Dye [14]. The projective-uniqueness and incompleteness statements used here appear in Storme and Van Maldeghem [25]. In the later terminology of Blokhuis, Seress, and Wilbrink, the same configuration is a complete exterior set of a conic [2]; their broader sets-without-tangents work and Van de Voorde's account give the adjacent literature [1, 26]. The six-arc census in $\mathrm{PG}(2,11)$ is due to Sadeh [18, 22]; see also the tables in Hirschfeld's book [17]. We recompute its extension-count spectrum as a byproduct of the rigidity theorem's exhaustive search and claim no priority on the raw numbers. The identification of this code's projective deep-hole syndrome locus with the A_5-invariant conic (Proposition 3.1) is recorded, with a kernel-checked certificate, in the companion arcs paper [5, Prop. 8.7]; we reprove it here to keep the present paper self-contained and claim no priority for it either. What we claim as new is the *reading* of that geometry: that the conic-containment condition characterizes the Clebsch hexagon among all six-arcs and recovers A_5 from a purely coding-theoretic hypothesis, not by construction; the global and local quantitative gap theorem accompanying that rigidity; the invariant chirality bipartition of the weight-three support types and its induced per-coset leader split; and the unconditional finite-field argument isolating $q = 11$.

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C}:\ XZ = Y^2$$

in $\mathrm{PG}(2,11)$, with its 12 points identified with $\mathrm{PG}(1,11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}(3,11)$ is $\mathrm{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

Definition 2.1 (Clebsch hexagon)**.** Let $A_5 \subset \mathrm{PGL}_2(11)$ be an icosahedral subgroup. The *Clebsch hexagon* is the six-point orbit of poles of the six 5-fold-axis chords of A_5 acting on $\mathcal{C}$; equivalently, the six points of $\mathrm{PG}(2,11)$ each of whose two tangency lines to $\mathcal{C}$ meet $\mathcal{C}$ in an antipodal pair fixed by a Sylow 5-subgroup of A_5.

For a nonsingular conic over an odd field, an *external point* lies on two tangents, while an *external line* is disjoint from the conic. Edge constructs exactly this $q = 11$ hexagon: its six vertices are external points, all fifteen joins are external lines, and the joins concur three at a time at ten internal Brianchon points [4, §§29–32]. In the terminology of Blokhuis, Seress, and Wilbrink, it is the six-point complete exterior set of $\mathcal{C}$ [2]; see Van de Voorde [26] for the later exterior-set context. This exterior-set condition says only that the joins avoid $\mathcal{C}$. Proposition 3.1 is the strictly stronger covering statement that those joins cover every point outside the hexagon and $\mathcal{C}$. Van de Voorde also relates the broader sets-without-tangents problem to stopping sets in projective-plane LDPC codes; that is a different coding connection from the MDS covering-radius and deep-hole interpretation developed here.

There is also a more direct fixed-conic binary-code lineage. Droms, Mellinger, and Meyer formed LDPC codes from the four internal/external and passant/secant point–line incidence matrices [15]; two of their dimension conjectures—the secant–external and passant–internal cases—were subsequently proved in [23, 20]. Wu then used whole $\mathrm{PSL}_2(q)$-orbit families of conics contained in the internal-point set, and Madison and Wu treated the external-point analogue [27, 21]. These are binary incidence codes whose coordinates range over an entire internal or external point class. For example, at $q = 11$ the passant–internal null-space code has parameters $[55, 25]_2$, while Wu's internal-orbit-conic null-space code has parameters $[55, 31]_2$. They are therefore different code objects from the six-coordinate $\mathbb{F}_{11}$-linear MDS code studied here; the shared conic, polarity, and $\mathrm{PSL}_2(11)$ geometry is infrastructure rather than a competing covering-radius or deep-hole identification.

This is projectively the arc K_2 of Storme and Van Maldeghem [25, Prop. 11] at $q = 11$, and the hexagon studied synthetically by Dye [14] under the existence condition 5 a square in the base field. By [25, Prop. 12] it is the unique $\mathrm{PGL}(3,11)$-orbit of six-arcs with stabilizer A_5; by Dye's classification its five self-polar triangles are self-polar with respect to the same conic $\mathcal{C}$, and it lies on no conic.[1] The arcs of $\mathrm{PG}(n,q)$ fixed by a *primitive* A_5 or A_6 were catalogued more generally by O'Keefe and Storme [19], of which the present arc is the planar A_5 case (its stabilizer is 2-transitive on the six points, hence primitive); we cite that catalogue for completeness, and take the planar uniqueness statement we use from [25, Prop. 12]. Storme and Van Maldeghem's Proposition 13 records, by computer search, that this six-arc is incomplete at $q = 11$ (unlike at $q = 9$, where the analogous A_5-hexagon is complete); equivalently, its associated redundancy-three code has covering radius exactly three. Incompleteness gives the lower bound, and the three-dimensional syndrome space gives the upper bound.

We work throughout with the explicit representative

$$A = \{(1,10,0),\ (1,9,1),\ (1,4,7),\ (1,8,5),\ (0,1,4),\ (1,1,7)\},$$

already used in the companion arcs manuscript [5] for the same reasons: it is disjoint from $\mathcal{C}$, no three of its points are collinear, and its stabilizer in $\mathrm{PGL}(3,11)$ has order 60.

3 The code and its projective syndrome locus

Form the 3×6 matrix H whose columns are (representatives of) the six points of A, and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^{\mathsf{T}} = 0\}$$

be the resulting $[6,3,4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [24].

By the arc–coset dictionary [7, Def. 6.1, Thm. 6.3], for every nonzero syndrome s above a projective direction $x = [s]$, the corresponding coset has minimum weight one, two, or three according as $x \in A$, x lies on a secant of A, or x lies on neither. In the last case each of the $q-1$ cosets above x has exactly $\binom{n}{3}$ weight-three leaders, here $\binom{6}{3} = 20$. Their Theorem 6.3 is stated for an arbitrary n-arc of $\mathrm{PG}(2,q)$ and its associated $[n, n-3, 4]_q$ MDS code, with no Reed–Solomon, normal-rational-curve, or conic hypothesis and no restriction on q or the characteristic; it therefore applies to the non-GRS code C at hand. (The GRS and conic specializations appear separately in their Theorems 6.4, 6.5, 6.8, and 6.10, which we do not use. The leader count is also the $d = 4$, $n = 6$ case of the general $\binom{n}{d-1}$ of [7, Thm. 7.7], so it is an instantiation of a uniform formula rather than a coincidence at these parameters.) Finally, by their Definition 6.1 together with the definition of arc completeness, adjoining x as a seventh arc column preserves the MDS property exactly when x is missed by every secant.

Proposition 3.1 (The projective deep-hole syndrome locus is the conic)**.** *The covering radius of C is three, and its projective deep-hole syndrome locus $\mathcal{D}_{\mathrm{proj}}(C)$ — the set of directions $x \notin A$ on no secant of A — is exactly the twelve points of $\mathcal{C}$.*

[1]The covering statement does not occur in Edge's $q = 11$ construction or in Van de Voorde's accessible account of exterior sets, and it is not among the stated results in Dye's author-written zbMATH summary (Zbl 0698.20032) or his 1997 self-recap. The two original Blokhuis–Seress–Wilbrink papers remain the priority check to be completed before submission; until then, our priority claim for the covering statement is conditional.

Computer-assisted proof. A direct finite computation over the 133 points of $\mathrm{PG}(2,11)$: for each of the $\binom{6}{2} = 15$ secants of A, mark the $q-1 = 10$ points it carries other than its two arc points. The marked set is $\mathrm{PG}(2,11) \setminus (A \cup \mathcal{C})$ — all 115 points off the arc except the twelve points of $\mathcal{C}$, each of which has secant index zero. By the dictionary above, this uncovered locus is exactly the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three rather than two. □

The computation is reproduced independently by `check_rigidity_degenerate_conic.py` as the $|\mathcal{U}(A)| = 12$ entry of the census of Section 4.

This identification, together with its coset data and a kernel-checked Lean certificate (root module `lean/RelativeConicArcs/Q11Coding.lean`), is also recorded in the companion arcs paper [5, Prop. 8.7], where it arises as one instance of that paper's secant-index spectrum; we restate it here, with a self-contained proof, because it is the starting point of the rigidity theorem that is the present paper's subject. The results of Sections 4–6 — the rigidity equivalence, the gap theorem, the chirality invariant, and the isolation of $q = 11$ — do not appear in that companion paper; their broader priority is treated separately in the literature comparison of Section 4.

Corollary 3.2 (Directions, cosets, leaders, and received words)**.** *The projective deep-hole syndrome locus has* 12 *directions, exactly the points of* $\mathcal{C}$*. Above them lie* $12(11-1) = 120$ *nonzero maximum-distance syndromes, hence* 120 *deep-hole cosets. Each such coset has* $\binom{6}{3} = 20$ *minimum-weight leaders, for* 2400 *leaders in all, and contains* $|C| = 11^3 = 1331$ *received words. Consequently* C *has* $120 \cdot 1331 = 159720$ *received-word deep holes.*

Proof. Immediate from Proposition 3.1, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \to \mathrm{PG}(2,11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. □

Proposition 3.3 (One orbit on received-word deep holes)**.** *The monomial automorphism group of* C *is transitive on the* 120 *deep-hole cosets. Let* Γ *be the group generated by these automorphisms and by translations* $v \mapsto v + c$ *for* $c \in C$*. Then*

$$|\Gamma| = |C|\,|\mathrm{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ *is transitive on all* 159720 *received-word deep holes. The stabilizer of each such word is cyclic of order five.*

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six axis chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^\times$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w, choose $m \in \mathrm{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w. Both kinds of generator are Hamming isometries preserving C. The monomial group normalizes the translation subgroup, since $m\,t_c\,m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \mathrm{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. □

As an independent finite verification, `check_chirality.py` constructs $C = \ker H$, verifies all $600 \cdot 1331$ monomial images of its codewords, and certifies the resulting 159720-element orbit.

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight $0, 1, 2, 3$, summing to $11^3 = 1331$); in particular the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters; we record it as a sanity check, not a novelty.

3.1 Certified structural facts

The following facts are either forced by the MDS parameters or are direct consequences of the arc's known geometry; we record them because they anchor the code's symmetry and design-theoretic reading, not as new results.

- **Automorphism groups.** The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the six coordinate supports in its exotic degree-six action. It is the support-permutation quotient of the monomial automorphism group, which fits into the central exact sequence
$$1 \longrightarrow \mathbb{F}_{11}^{\times} \longrightarrow \mathrm{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$
The kernel consists of the ten global coordinate scalars. The sequence splits: since $\gcd(3, 10) = 1$, every class in $\mathrm{PGL}(3, 11)$ has a unique determinant-one representative, and these representatives multiply; their associated monomial lifts give an A_5 complement. Thus $\mathrm{MAut}(C) \cong C_{10} \times A_5$ has order 600. By contrast, for the displayed choice of column representatives the subgroup of pure coordinate permutations is trivial. The exact support image, scalar kernel, monomial order, pure subgroup, and transitivity on the 120 nonzero syndromes above $\mathcal{C}$ are independently checked by `check_code_automorphisms.py`; Proposition 3.1 identifies these with the maximum-distance syndromes.

- **Design view.** The MDS parameters make C an orthogonal array $\mathrm{OA}(11^3, 6, 11, 3)$ of strength three; this is an automatic consequence of the $[6, 3, 4]$ parameters and is recorded here only to connect the code to the design-theoretic framing this venue expects.

- **Five self-polar triangles.** Edge's finite-plane construction [4, §§30.2–31] and Dye's general-field synthetic theory [14] exhibit five triangles on the six vertices, each self-polar with respect to the unique conic $\mathcal{C}$ stabilized by the hexagon's projective stabilizer — the same conic as the projective deep-hole syndrome locus of Proposition 3.1. In the six-coordinate labelling, the three vertex-pairs underlying each triangle form a perfect matching, and the five matchings partition all fifteen pairs: they are a synthematic total. Pairing two of these triangles produces an alternating six-cycle. Its two colour classes form a complementary pair of three-element coordinate supports, while its opposite-vertex matching gives one of the ten Brianchon concurrences. Thus the ten unordered pairs of triangles index simultaneously the ten Brianchon points and the ten complementary support pairs; this is the geometric origin of the Petersen graph made precise in Proposition 5.1.

4 The rigidity theorem

The content of Proposition 3.1 is, on its own, a single finite computation for one known arc. The following theorem is the paper's main claim: among *all* six-arcs of $\mathrm{PG}(2, 11)$, the property “the projective deep-hole syndrome locus lies on a conic” picks out the Clebsch hexagon uniquely, and does so in a way that recovers its projective stabilizer from the coding condition rather than assuming it.

Remark 4.1 (Two conditions, one word apart). The condition studied here is that $\mathcal{U}(A)$ — the projective uncovered, or projective deep-hole syndrome, locus — lies on a conic. It is *not* the familiar condition that the arc A itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are logically unrelated: they constrain different point sets, and the arcs satisfying them are disjoint. Indeed every six-arc lying *on* a conic in $\mathrm{PG}(2,11)$ has $|\mathcal{U}(A)| \in \{18,19,20,22\}$, so by (iii) none of them comes anywhere near the condition of the theorem.

Census framing. The projective-equivalence classes of k-arcs in $\mathrm{PG}(2,11)$ were classified by Sadeh [22, 18]; see also Hirschfeld [17, Ch. 14]. Over the six-arc classes of that census we compute, for each representative A, the projective uncovered locus $\mathcal{U}(A)$ — the points off A and off all fifteen of its secants, equivalently the points extending A to a projective seven-arc — and read $\mathcal{U}(A)$ as the projective covering-radius-three syndrome data of the associated $[6,3,4]_{11}$ code. *We claim no priority on the six-arc census, nor on the raw extension-count data* $|\mathcal{U}(A)|$ *that an extension-based classification entails*: the size of $\mathcal{U}(A)$ is a standard projective invariant of any arc, and the census itself is Sadeh's. What we contribute is the geometric and coding-theoretic meaning of $\mathcal{U}(A)$: that its containment in a conic singles out the Clebsch hexagon and forces A_5, that this containment is rigid rather than merely typical, and that for the singled-out class the complete projective deep-hole syndrome locus is the full set of $\mathbb{F}_{11}$-points of a conic.

Theorem 4.2 (Rigidity theorem). *For a six-arc $A \subset \mathrm{PG}(2,11)$, the following are equivalent:*

(i) the projective deep-hole syndrome locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);

(ii) $\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$-points of a nonsingular conic;

(iii) $|\mathcal{U}(A)| \leq 15$ (in fact, $|\mathcal{U}(A)| = 12$);

(iv) A is $\mathrm{PGL}(3,11)$-equivalent to the Clebsch hexagon;

(v) the stabilizer of A in $\mathrm{PGL}(3,11)$ contains A_5.

Computer-assisted proof. Every four points of a six-arc, no three collinear, form a projective frame; normalizing that frame to $e_1, e_2, e_3, (1,1,1)$ and sweeping the remaining two points over all legal completions meets every projective class of six-arc at least once, giving 1548 frame-normalized representatives. The enumeration is by representatives, not by classes: a class with stabilizer S appears $360/|S|$ times among them, so the class count is $\sum_{\mathrm{reps}} |S|/360 = 15$. Exhaustiveness is what the argument needs, and representatives supply it; the multiplicities are an artifact of the normalization and carry no projective meaning. For each representative, $\mathcal{U}(A)$ and its conic membership are computed by exact linear algebra (secant coverage by direct enumeration; conic containment by the nullspace of the 6×6 quadratic-monomial evaluation matrix restricted to $\mathcal{U}(A)$, allowing degenerate quadrics). This yields the extension-count histogram

$$|\mathcal{U}(A)| \in \{12,16,18,19,20,21,22\}, \qquad \text{with multiplicities } \{6,30,150,300,630,360,72\}$$

over the 1548 normalized representatives (summing to 1548); the multiplicities are an artifact of the frame-normalized enumeration, and the projective invariant is the value-set itself together with its per-class assignment, which we grant outright to the Hirschfeld–Sadeh classification. The minimum, $|\mathcal{U}(A)| = 12$, is attained by exactly six of the 1548 representatives, all in a single $\mathrm{PGL}(3, 11)$-orbit (multiplicity $6 = 360/60$, consistent with a stabilizer of order 60); this orbit is the Clebsch hexagon class, and on it $\mathcal{U}(A)$ is exactly the twelve points of the A_5-invariant conic $\mathcal{C}$. No other class has $\mathcal{U}(A)$ contained in *any* conic — every one of the remaining classes, including the concyclic six-arcs (those whose own six points lie on a conic, 252 of the 1548 representatives), has $|\mathcal{U}(A)| \geq 16$ with $\mathcal{U}(A)$ off every conic. This gives (iii)⇔(iv)⇔(v), and (ii)⇒(i) is immediate. On that concyclic slice there is a useful independent check: if $t(H)$ counts concurrent triples among the fifteen chords of H, with multiplicity and including the sixty triples forced at H, then the six-arc chord–extension identity gives $t(H) + |\mathcal{U}(H)| = 82$: indeed, Lemma 6.2 gives $|\mathcal{U}(H)| = 22 - c(H)$, while $t(H) = 60 + c(H)$. Thus the four on-conic values $|\mathcal{U}(H)| \in \{18, 19, 20, 22\}$ correspond to $t(H) \in \{64, 63, 62, 60\}$. We use only this counting identity here; the separate Mathieu-hexad characterization of its extremal stratum is not part of the present paper.

For (i)⇒(ii) we exclude the degenerate case, since condition (i) as stated permits a degenerate conic (a line-pair or double line): the conic-containment test above is the rank over $\mathbb{F}_{11}$ of the $|\mathcal{U}(A)| \times 6$ quadratic-monomial evaluation matrix, which detects *any* nonzero quadratic form vanishing on $\mathcal{U}(A)$, degenerate or not. The exhaustive computation shows that exactly the six Clebsch-class representatives have this rank below 6; for each, the kernel is one-dimensional and its quadratic form is nonsingular (nonzero 3×3 determinant), with $\mathcal{U}(A)$ equal to all twelve $\mathbb{F}_{11}$-points of that conic; and *no* six-arc has $\mathcal{U}(A)$ supported on a degenerate conic. Hence (i)⇒(ii), completing the equivalence. □

The standalone artifact `check_rigidity_degenerate_conic.py` implements the exhaustive computation just described, reproduces the displayed histogram, and returns exactly the six normalized Clebsch-class representatives whose extension loci lie on conics; every containing conic is nonsingular.

Corollary 4.3 (Monomial characterization)**.** *Up to monomial equivalence, C is the unique linear $[6, 3, 4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus lies on a conic. In particular, A_5 is recovered from this coding condition, not assumed as a hypothesis.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A, and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Theorem 4.2 makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. The converse is immediate, and the stabilizer statement is clause (v) of the theorem. □

Remark 4.4 (Which implications are new)*.* The equivalence (iv)⇔(v) is not ours and is not new: that the Clebsch hexagon is the unique $\mathrm{PGL}(3, 11)$-orbit of six-arcs whose stabilizer contains A_5 is Storme and Van Maldeghem [25, Prop. 12], quoted in Section 2 above, and the classification of six-arcs that underlies it is Sadeh's. The content of the theorem is the implication (i)⇒(iv), and with it the equivalence of the projective syndrome conditions (i)–(iii) with the classical characterizations (iv)–(v). It is that bridge — from a covering-radius property of the code to a projective classification — that is the claim, and the corollary above is its consequence.

The rigidity theorem has three quantitative refinements: a global gap in the extension-count census, a projectively invariant distance from the nearest conic, and a local gap under single-vertex replacement. The first is a statement of the standard extremal shape — there is no object strictly between the extremal value and the next, and the objects attaining the extremal value are classified — familiar from the minimum-side gap results for arcs and blocking sets, such as Blokhuis–Bruen's [12] theorem on the sizes realized just above the minimum. Here the invariant is the projective deep-hole-direction count $|\mathcal{U}(A)|$.

For the second, let $\mathfrak{C}$ be the set of nonsingular conics in $\mathrm{PG}(2,11)$ and define the *nearest-conic discrepancy*

$$\delta(A) = \min_{\mathcal{Q}\in\mathfrak{C}} |\mathcal{U}(A) \triangle \mathcal{Q}(\mathbb{F}_{11})|.$$

This is a PGL-invariant function, not a metric on arcs: equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A)$, while PGL permutes $\mathfrak{C}$.

For the local statement, let $\mathcal{G}_1$ be the graph whose vertices are embedded six-arcs in $\mathrm{PG}(2,11)$, with two vertices adjacent when their point sets have symmetric difference two. Fix the displayed Clebsch arc A and its standard conic $\mathcal{C}$, and put

$$d_{\mathcal{C}}(B) = |\mathcal{U}(B) \triangle \mathcal{C}|$$

for every embedded six-arc B; clause (c) restricts this discrepancy to the neighbors of A.

Theorem 4.5 (Gap theorem)**.** *The Clebsch configuration is rigid rather than merely stable.*

(a) (Census gap.) *There is no six-arc $A \subset \mathrm{PG}(2,11)$ with $12 < |\mathcal{U}(A)| < 16$. Furthermore, every six-arc attaining $|\mathcal{U}(A)| = 12$ is $\mathrm{PGL}(3,11)$-equivalent to the Clebsch hexagon, and every six-arc with $|\mathcal{U}(A)| \geq 16$ has $\mathcal{U}(A)$ contained in no conic.*

(b) (Global nearest-conic gap.) *On the fifteen $\mathrm{PGL}(3,11)$-classes of six-arcs, the exact class-counted histogram of δ is*

$$\{0{:}1,\ 12{:}2,\ 13{:}1,\ 14{:}3,\ 15{:}1,\ 16{:}4,\ 17{:}2,\ 18{:}1\}.$$

Its unique zero is the Clebsch class. Thus every non-Clebsch class has $\delta(A) \geq 12$, and the bound is sharp.

(c) (Perturbation gap.) *Each deleted vertex of A has exactly 42 legal replacements, giving 252 distinct neighbors B. Their exact fixed-conic discrepancy histogram is*

$$\{18{:}30,\ 19{:}60,\ 20{:}90,\ 22{:}42,\ 24{:}30\}.$$

The corresponding histogram of $|\mathcal{U}(B) \cap \mathcal{C}|$ is $\{4{:}60, 6{:}132, 7{:}60\}$, so at most seven conic directions remain; no neighbor has $\mathcal{U}(B)$ contained in any conic, degenerate or otherwise. Under $\mathrm{Stab}(A) \cong A_5$, the neighbors split into eight orbits whose pairs (orbit size, fixed-conic discrepancy) are

$$(30,18), (60,19), (30,20), (30,20), (30,20), (12,22), (30,22), (30,24).$$

Thus every vertex adjacent to A in $\mathcal{G}_1$ has $d_{\mathcal{C}}(B) \geq 18$.

Remark 4.6 (Global versus fixed-conic discrepancy). The closing statement is local to the fixed pair $(A, \mathcal{C})$ and is not a global nearest-arc theorem. The conic-preserving projectivity $g : (X,Y,Z) \mapsto (4X, 2Y, Z)$ sends A to a distinct embedded six-arc, while equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A) = \mathcal{C}$. Hence $d_{\mathcal{C}}(gA) = 0$, although $gA \neq A$. This is exactly why the global statement minimizes over conics and passes to projective classes: gA remains in the Clebsch class and has $\delta(gA) = 0$. The lower bound eighteen is specifically for the neighbors of the fixed embedded A in $\mathcal{G}_1$; the global non-Clebsch lower bound is the separate value twelve.

Remark 4.7 (Two spectra, one word apart). Clause (a) is a gap in the spectrum of the *projective deep-hole- direction count* $|\mathcal{U}(A)|$ at fixed arc size $k = 6$. It is not a statement about the much-studied spectrum of the *sizes* k for which a complete k-arc exists, which at $q = 11$ was determined by Faina, Marcugini, Milani and Pambianco [13]: there the invariant is k itself and the relevant constants are $t(2, 11) = 7$ and $m''(2, 11) = 10$. The two meet only at the boundary $\mathcal{U}(A) = \emptyset$, where an arc is complete; that $t(2, 11) = 7$ exceeds six is exactly why no six-arc of $\mathrm{PG}(2, 11)$ is complete, and hence why the histogram above has no zero bin.

Computer-assisted proof. The first clause restates the exhaustive classification of Theorem 4.2. For the second, `check_global_conic_gap.py` normalizes all 360 ordered four-frames of each of the 1548 representatives to obtain exactly fifteen projective-class keys. Independently it enumerates all 177156 projective ternary quadrics, retains exactly 160930 nonsingular conics, and computes the minimum symmetric difference for each class. It asserts the displayed class histogram, a direct-XOR/intersection-formula cross-check, and the unique zero class.

For the third clause, the local checker enumerates the 133 projective points, deletes each of the six vertices in turn, and tests every replacement outside A for the six-arc condition. It asserts the six per-vertex counts, distinctness of the 252 resulting arcs, both displayed histograms, and conic noncontainment by exact quadratic-monomial rank. It then uses the sixty exact projectivities in the stabilizer to certify the eight closed orbits and recover both histograms from their rows. It also verifies the projectivity in the preceding remark and directly recomputes $\mathcal{U}(gA) = \mathcal{C}$. Its source is `check_perturbation_gap.py`. $\square$

All three clauses are exhaustive finite checks over $\mathrm{PG}(2, 11)$; the three standalone paper-package checkers expose their exact assertions and exit nonzero on any mismatch.

Remark 4.8 (Why these facts are not already recorded). The locus $\mathcal{U}(A)$ is not a new object: by Segre's theory of the tangent envelope of an arc, a point extends A exactly when its pencil is a component of that envelope, so $\mathcal{U}(A)$ is the classical extension locus — see Ball and Lavrauw [8, Thms. 39–41] for the statement in both parities. Two separate reasons account for the absence of the facts above from that literature, and neither is that the objects were unknown. First, the arc-classification line has long had the census we use — the six-arc classes of $\mathrm{PG}(2, 11)$ are Sadeh's — but no motive to compute $\mathcal{U}(A)$ on it: its purpose there is the construction of cubic surfaces, which consumes only the condition that A itself lie on no conic. Second, the arc-extension line has exactly this motive, but its theorems are large-k theorems whose hypotheses exclude our case: over $\mathbb{F}_q$ with q odd the envelope results apply from $|A| \geq 2q/3 + 2$, which is ten points at $q = 11$, and Segre's extension bound needs $k > q - \sqrt{q}/4 + 25/16 \approx 11.73$, against our $k = 6$. At $k = 6$ the envelope has class $14 > 12$, so membership in it constrains nothing. The interest of the statements above therefore rests not on their inaccessibility but on the coding reading of $\mathcal{U}(A)$ and on the rigidity theorem that the gap protects.

5 The chirality bipartition

The six columns of A give $\binom{6}{3} = 20$ three-element subsets. The stabilizer A_5 has two orbits of size ten on them, interchanged by complementation: for every S, the orbit of S never contains S^{c}. The triples also form ten complementary pairs $\{S, S^{\mathsf{c}}\}$. Choose the member in either one of the two A_5 orbits consistently from every pair and join two pairs when their chosen representatives share exactly two columns. The resulting graph is the

Petersen graph; choosing the other orbit gives the same adjacency because simultaneous complementation preserves intersection sizes.

There is a direct geometric labelling of these ten vertices.

Proposition 5.1 (Brianchon–support dictionary). *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [4, 14], regarded as the five perfect matchings in their synthematic total, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T,T')$ be the perfect matching joining opposite vertices of this cycle, and let $\beta(T,T') = \{S, S^{\mathsf{c}}\}$ be its unordered bipartition.*

The three chords indexed by $M(T,T')$ concur at a point $b(T,T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \qquad \{T,T'\} \longmapsto b(T,T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S,S^{\mathsf{c}}\} : |S| = 3\}, \qquad \{T,T'\} \longmapsto \beta(T,T'),$$

are A_5-equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of the synthematic total. Hence $\{T,T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T,T')$, so the ten triangle pairs give exactly the ten perfect matchings outside the total. Edge's Clebsch construction says that the three chords of each such matching concur, at the ten distinct Brianchon points [4, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above. All concurrences, the ten multiplicity-three and fifteen multiplicity-one intersections of disjoint chords, and the equivariance and support-map compatibility follow from these two equivariant bijections. □

As an independent verification, `check_chirality.py` checks all chord intersections, the exact $3^{10}1^{15}$ multiplicity ledger, both equivariance statements, and compatibility with the support map.

Edge's two systems of eleven Clebsch hexagons over a fixed conic, exchanged by the operations outside $\Omega^+(3,11)$ [4, §§29,32], are a classical antecedent of the same two-class motif. They are classes of hexagons, however, not the support-orbit bipartition proved below.

Proposition 5.2 (Invariant support chirality). *The twenty three-element coordinate supports split into two complementary A_5-orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s, each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the* 2400 *minimum-weight leaders split as* $1200 + 1200$.

Every monomial automorphism of C preserves this unordered bipartition. The other coset in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. Thus the intrinsic structure is a two-class decomposition, or an unbased $\mathbb{Z}/2$-torsor, rather than a canonically oriented $\mathbb{Z}/2$-valued label. By Proposition 5.1, the ten complementary support pairs are canonically both the two-subsets of the five self-polar triangles and the ten Brianchon points.

Proof. The two size-ten support orbits and the normalizer action follow by applying the explicit exotic degree-six A_5 to the $\binom{6}{3} = 20$ subsets. Of the six synthematic totals on the six coordinates, exactly one is stabilized by this A_5 (indeed by its full S_5 normalizer); it is the total supplied by the five triangles. Any two of its perfect matchings are edge-disjoint, so their union is a connected 2-regular bipartite graph on six vertices, hence a six-cycle with a unique unordered $3+3$ bipartition. The ten pairs of matchings give all ten complementary support pairs. This construction is equivariant under the full normalizer, and direct comparison shows that disjoint pairs of triangles give exactly the intersection-two adjacency defined above. Fix a maximum-distance syndrome s and a three-support S. The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s. None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A, contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S. Finally, every monomial automorphism induces a support permutation in A_5, by the exact sequence in Section 3.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits, but the pure-permutation audit there and the monomial support-image audit show that it does not preserve C, even after coordinate rescaling. $\square$

The complete finite ledger in the preceding proof is independently checked by `check_chirality.py`, including the support orbits, Petersen adjacency, all perfect matchings and synthematic totals, the triangle-pair/support/Brianchon dictionary, both equivariance statements, the full normalizer, all $120 \cdot 20$ coefficient-bearing leaders, and all $600 \cdot 2400$ monomial-action cases.

The bipartition is therefore intrinsic to the code's covering-radius data, but its two sides have no preferred signs without an additional orientation.

6 Why $q = 11$

The conic-filling syndrome phenomenon of Sections 3–4 is specific to $q = 11$, even without a symmetry hypothesis on the arc. A universal chord-multiplicity identity makes this a four-field question, and the only nontrivial residue is a classical exact-distance problem in the Sylvester graph.

Lemma 6.1 (Secant-covering bound)**.** *Let A be a six-arc of $\mathrm{PG}(2,q)$ disjoint from a non-singular conic $\mathcal{C}$, and suppose A is complete outside $\mathcal{C}$: every point of $\mathrm{PG}(2,q)$ lying off A and off $\mathcal{C}$ lies on some secant of A. Then*

$$15(q-1) \;\geq\; q^2 - 6,$$

equivalently $q^2 - 15q + 9 \leq 0$, so $q \leq 14$.

Proof. Since A has six points and $\mathcal{C}$ has $q+1$ points, and A is disjoint from $\mathcal{C}$,

$$\left|\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})\right| = (q^2+q+1) - 6 - (q+1) = q^2 - 6.$$

Each of the $\binom{6}{2} = 15$ secants of A is a line of $\mathrm{PG}(2,q)$, hence has $q+1$ points, exactly two of which lie in A; so each secant covers at most $q-1$ points of $\mathrm{PG}(2,q) \setminus A$, and in particular at most $q-1$ points of $\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})$. Completeness outside $\mathcal{C}$ says the fifteen secants together cover all $q^2 - 6$ points of this set, so

$$q^2 - 6 \;\leq\; 15(q-1).$$

(No accounting for overlap between secants is needed here: the bound only uses that fifteen sets of size at most $q-1$ cover a set of size q^2-6, which requires the sum of their sizes to be at least q^2-6 regardless of how much they overlap; overlap can only make the true covered count smaller than the sum, never larger, so the inequality direction needed for an upper bound on q is exactly the one overlap cannot violate.) Rearranging, $q^2-15q+9 \le 0$, whose real roots are $\frac{15\pm\sqrt{189}}{2} \approx 0.62, 14.38$; since q is a positive integer, $q \le 14$. $\square$

Lemma 6.2 (Chord-defect identity)**.** *Let A be a six-arc in* $\mathrm{PG}(2,q)$*, and let $c(A)$ be the number of points off A incident with three of its fifteen chords. Then*

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \qquad 0 \le c(A) \le 15.$$

Consequently, if $|\mathcal{U}(A)| = q+1$, then

$$c(A) = (q-6)(q-9).$$

Proof. At a point off A, concurrent chords have pairwise disjoint endpoint pairs, so at most three chords occur. Let n_i count the off-arc points on exactly i chords. Counting chord–point incidences and pairs of chords with disjoint endpoints gives

$$n_1 + 2n_2 + 3n_3 = 15(q-1), \qquad n_2 + 3n_3 = \binom{15}{2} - 6\binom{5}{2} = 45.$$

Writing $c(A) = n_3$, the number of covered points off A is $n_1 + n_2 + n_3 = 15q - 60 + c(A)$. Subtracting this from the q^2+q-5 points off A proves the identity. Each triple concurrence uses one of the fifteen perfect matchings of the six endpoints, and a fixed matching can be concurrent at only one point; hence $c(A) \le 15$. The last display follows on setting $|\mathcal{U}(A)| = q+1$. $\square$

Theorem 6.3 (Uniqueness of $q = 11$)**.** *Let q be a prime power. If a six-arc $A \subset \mathrm{PG}(2,q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. Moreover, at $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Lemma 6.2 gives $c(A) = (q-6)(q-9)$ with $0 \le c(A) \le 15$. The prime-power condition therefore leaves only $q \in \{4, 5, 9, 11\}$: the values at 7 and 8 are negative, while every integer $q \ge 12$ gives $c(A) \ge 18$.

At $q = 4$, every six-arc is a hyperoval. Each point off it lies on three secants (count the six hyperoval points on the five lines through the point), so $\mathcal{U}(A)$ is empty. At $q = 5$, every six-arc is an oval and hence a nonsingular conic; the standard internal/external point counts show that every off-conic point lies on a secant, so again $\mathcal{U}(A)$ is empty.

It remains to exclude $q = 9$. Relative to the hypothetical conic $\mathcal{C}$, every chord of A is passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external; hence the five chords through each vertex force all six vertices of A to be internal. Let H be the graph on the 36 internal points, with $P \sim Q$ when $Q \in P^\perp$ under the conic polarity. Standard conic counts give the intersection array

$$\{5, 4, 2; 1, 1, 4\},$$

so H is the Sylvester graph [10, §13.1.2]; see also the combinatorial uniqueness proof in [11, Thm. 6]. For distinct internal points, the unique possible common neighbor is $P^\perp \cap Q^\perp = (PQ)^\perp$; consequently PQ is passant exactly when P, Q are at distance two in H. Thus A would be a six-clique in the exact-distance-two graph of H. But [9, Table 1] gives $\mathrm{eq}_2(H) = 5$, a contradiction.

Therefore $q = 11$. The final assertion follows from Theorem 4.2, which puts every matching $q = 11$ representative in the single Clebsch orbit. $\square$

The smaller from-scratch certificate `../../notes/2026-07-15-c181-sylvester-q9.py` independently constructs $\mathbb{F}_9$, $\mathrm{PG}(2,9)$, the polarity graph and its full intersection array, verifies passant join if and only if distance two for all internal-point pairs, and computes the exact clique number 5. The all-field checker `check_small_q_uniqueness.py` supplies the separate census below.

For independent verification, every six-arc contains an ordered projective four-frame, which may be normalized to $(1,0,0),(0,1,0),(0,0,1),(1,1,1)$. Enumerating the remaining unordered pair with exact field arithmetic gives the following census; exponents in the third column record multiplicities.

q	representatives	histogram of $\lvert\mathcal{U}(A)\rvert$	conic matches
2	0	—	0
3	0	—	0
4	1	0^1	0
5	3	0^3	0
7	70	$0^{40}, 2^{30}$	0
8	195	$0^{45}, 4^{150}$	0
9	441	$0^6, 4^{15}, 6^{120}, 7^{120}, 8^{180}$	0
11	1548	$12^6, 16^{30}, 18^{150}, 19^{300}, 20^{630}, 21^{360}, 22^{72}$	6
13	4015	$36^{85}, 38^{210}, 39^{480}, 40^{1080}, 41^{1800}, 42^{360}$	0

The enumeration uses exact field arithmetic, including explicit asserted tables for $\mathbb{F}_4, \mathbb{F}_8$, and $\mathbb{F}_9$; its conic test is also exercised on nonsingular and singular forms in characteristic two. It is reproduced by `check_small_q_uniqueness.py` and agrees with the conceptual proof above.

The theorem is an unconditional field-size classification with a conceptual small-field proof. Within the icosahedral construction, three further features meet at 11: $p+1=12$, so the twelve vertices of the icosahedron exhaust the projective line; p is prime to 60, so the icosahedral group survives reduction faithfully; and the fifteen secants of a six-arc have enough capacity to cover every point off the conic. The failure of the same symmetric recipe at the next admissible prime is nevertheless worth recording concretely. At $q=19$ ($19 \equiv -1 \pmod{10}$, so the rationality filter passes and only Lemma 6.1 excludes it) the same recipe — the poles of the six five-fold axes of an icosahedral $A_5 < \mathrm{PSL}_2(19)$ — does still produce a genuine six-arc disjoint from the conic, so the object exists; what fails is the covering. We built that arc and enumerated its projective deep-hole syndrome locus directly (`check_q19_nonexample.py`), obtaining

$$\lvert\mathcal{U}(A)\rvert = 140 = 20 + 120$$

against a conic of only $q+1=20$ points. The arithmetic differs from $q=11$ in a way worth noting: there $5 \mid q-1$, so an order-five element splits and fixes two rational points of $\mathcal{C}$, whereas at $q=19$ we have $5 \mid q+1$, so the order-five elements lie in a non-split torus and fix no rational conic point at all — their fixed pairs are conjugate over $\mathbb{F}_{361}$. The chord through such a pair is Galois-stable and hence still a rational line, so its pole is still a rational point and the arc is still defined; we verify this in the script rather than assume it.

The shape of the failure is instructive. The twenty conic directions remain in the projective deep-hole syndrome locus — $\mathcal{C} \subset \mathcal{U}(A)$ still — but they no longer exhaust $\mathcal{U}(A)$: 120 further points of $\mathrm{PG}(2,19)$ escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic (the quadratic-monomial evaluation matrix of $\mathcal{U}(A)$ has full rank six). The deficit itself is forced by Lemma 6.1: the fifteen secants have total capacity $15(q-1)=270$, while $q^2-6=355$

points lie off the arc and off the conic, so at least 85 points must escape, and 120 in fact do. (That the twelve — here twenty — conic points also remain uncovered is a separate fact, not something the lemma predicts.) At $q = 11$ the same capacity, 150, comfortably exceeds the 115 points needing cover, and the excess is spent on overlap. The phenomenon therefore degenerates immediately past $q = 11$ rather than persisting into a family: the conic remains inside the projective syndrome locus but no longer exhausts it.

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